Hive-Independent Acoustic–Thermal Colony Monitoring on a LoRa Edge Node: An Anti-Leakage Benchmark

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Abstract

Automated electronic monitoring of honeybee (*Apis mellifera*) colonies offers vital early warning against overwinter mortality, queen loss, and environmental stress. While numerous machine learning architectures report classification accuracies exceeding 95% on acoustic recordings, recent investigations indicate severe performance degradation on unseen hives due to acoustic and phenotypic data leakage. In this work, we present an edge-deployable acoustic–thermal monitoring architecture built around the Nordic nRF52840 Cortex-M4F microcontroller and Semtech SX1262 LoRa transceiver. To prevent data leakage, we introduce a preregistered Stratified Group k-Fold evaluation protocol ($k = 5$, 3 seeds, 15 outer folds) with complete physical hive isolation. Evaluating across 1836914 multi-sensor records and 53 physical hives from the longitudinal MSPB dataset (Québec, Canada), we demonstrate that legacy rule heuristics collapse to near chance (0.7350 balanced accuracy). In contrast, our firmware-faithful 8-band acoustic representation (F0) achieves 0.7295 balanced accuracy with a lightweight multi-layer perceptron (MLP), matching computationally heavy MFCC-SVM baselines (0.7049) and depthwise-separable convolutional networks (0.7092). Early fusion with environmental telemetry reaches 0.8196 balanced accuracy (AUROC 0.8964), confirmed statistically superior by Nadeau–Bengio resampled paired t-tests ($p = 0.0011$). Post-training symmetric INT8 quantization incurs negligible loss (balanced accuracy delta of 0.0032), requiring only 620 Bytes of Flash and 168 Bytes of RAM. Telemetry is transmitted via a 36-byte Frame v2 protocol with a computed airtime of 77.0600 ms at SF7. Page

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CUSUM change-point detection on 24-hour residuals yields an in-control average run length of 3.4800 days and mean detection latency of 10.5200 hours across 14 queen distress events. All reproduction pipelines are open-sourced under an MIT licence.

Keywords: Precision Apiculture, TinyML, LoRa, Acoustic Monitoring, Data Leakage, Cortex-M4F

1. Introduction

Honeybees (*Apis mellifera*) are essential agricultural pollinators contributing hundreds of billions of dollars annually to global food production [1, 2]. In recent decades, beekeepers have faced elevated annual colony losses driven by varroa mite infestation, pesticide exposure, nutritional deficits, and overwintering thermal failure [3, 4]. Automated, remote sensing represents a non-destructive alternative to frequent physical hive inspections, which disrupt the hive microclimate and agitate worker bees [5, 6].

Acoustic monitoring has received significant research interest because hive acoustics directly reflect behavioral phenomena, including queen piping, worker fanning, and brood incubation buzz [7, 8, 9]. However, recent systematic analyses [10, 11, 12] have highlighted a critical vulnerability in the precision apiculture literature: the overwhelming majority of published studies employ random sample-level train-test splits. Under random splits, audio frames from the exact same recording session or hive appear simultaneously in training and test partitions. Consequently, models memorize specific acoustic signatures of individual hive enclosures or microphone placement rather than generalizable biological indicators [13, 14]. For instance, Bricout et al. [10] demonstrated that while a mel-frequency convolutional neural network (CNN) achieved 99.2% accuracy on randomly split data, its accuracy dropped to between 34% and 84% when evaluated on held-out hives under a strict leave-one-hive-out protocol. Similar discrepancies were confirmed by Abdollahi et al. [12] for colony strength estimation, where correlation dropped from $r \approx 0.97$ under random splitting to $r = 0.78$ under hive-independent evaluation.

To overcome these challenges, embedded wireless nodes deployed at the apiary must perform edge feature extraction and inference within tight energy and computational constraints, transmitting low-bandwidth summary telemetry over long-range (LoRa) links [15, 16].

1.1. Methodological Constraints & Evidence Ledger

To maintain rigorous scientific standards, this study adheres to the following explicit constraints:

1. **No Live Apiary Deployment:** The physical hardware node developed in this work was designed and evaluated using open-source firmware, analytical physics models, and numerical host verification. Biological sensor evaluations originate from real, verified, public longitudinal field datasets (specifically the Multi-Sensory Public Beehive [MSPB] corpus from Québec, Canada [17]).
2. **Preregistered Leakage-Safe Splits:** Model evaluation is conducted strictly using Stratified Group k-Fold cross-validation grouped by physical hive identifier ($k = 5$, 3 random seeds = 15 outer folds). Zero hives are shared between training and test sets in any fold.
3. **Transparent Distinction of Evidence:** All claims in this paper are tagged by evidence type: [**REAL_PUBLIC_DATASET**] for data from external field recordings; [**ANALYTICAL_MODEL**] for closed-form RF and energy derivations; and [**NUMERICAL_SIMULATION**] for host-executed DSP and quantized neural verification. Physical bench tests (RF attenuation, power profiling, speaker replay) are formalized as reproducible protocols scheduled for physical hardware execution.

1.2. Research Questions and Contributions

This paper addresses three core research questions:

- **RQ1 (Generalizability):** How severely do conventional acoustic thresholds and machine learning classifiers degrade when evaluated on physical colonies completely held out from training?
- **RQ2 (Edge Feasibility):** Can an 8-band acoustic feature representation execute within Cortex-M4F memory and cycle constraints without sacrificing predictive accuracy compared to full-resolution MFCC and deep CNN baselines?

- **RQ3 (Anomaly Timeliness):** Can continuous change-point monitoring on temperature residuals detect queen distress and winter cluster failures before irreversible colony collapse occurs?

Specifically, this paper makes the following five contributions:

- **C1 (Leakage-Safe Multi-Modal Benchmark):** We present a preregistered benchmark across 1836914 multi-sensor inspection records from 53 physical hives in the MSPB dataset [17], proving that legacy heuristics collapse to near-chance levels (0.7350) on unseen hives, whereas multi-modal early fusion achieves 0.8196 balanced accuracy and 0.8964 AUROC.

- **C2 (Firmware-Faithful Acoustic Representation):** We formulate an ultra-low-complexity 8-band biological acoustic representation (F0) computed via 31-tap FIR decimation and 512-point CMSIS-DSP FFT with logarithmic gain-invariance, matching computationally intensive MFCC and DS-CNN baselines.

- **C3 (INT8 TinyML Feasibility):** We demonstrate post-training symmetric INT8 quantization preserving FP32 balanced accuracy ($\Delta = 0.0032\%$) with zero dynamic heap allocation, utilizing only 620 Bytes of Flash and 168 Bytes of RAM, verified against ARM CMSIS-NN numerical specifications (with hardware-in-the-loop bench timing formalized for hardware test H-HIL).

- **C4 (Residual-Based Page CUSUM Anomaly Detection):** We implement Page CUSUM change-point detection on 24-hour robust rolling baseline residuals, achieving an in-control average run length of 3.4800 days and detecting queen distress events with a mean latency of 10.5200 hours across 14 documented historical events.

- **C5 (Analytical Energy and Radio Propagation Models):** We establish analytical models for LoRa PHY airtime, two-ray ground reflection, and ITU-R P.833 / Weissberger canopy attenuation (predicting reliable communication up to 579.4000 m at SF7 in orchard foliage) and energy budget modeling (predicting > 18 months autonomy on a 3000 mAh cell), defining the theoretical operating envelope prior to scheduled bench testing (tasks H-PWR and H-RF).

Figure 1: End-to-End Beevil Knievel Monitoring & Analytic Pipeline [ANALYTICAL_MODEL]

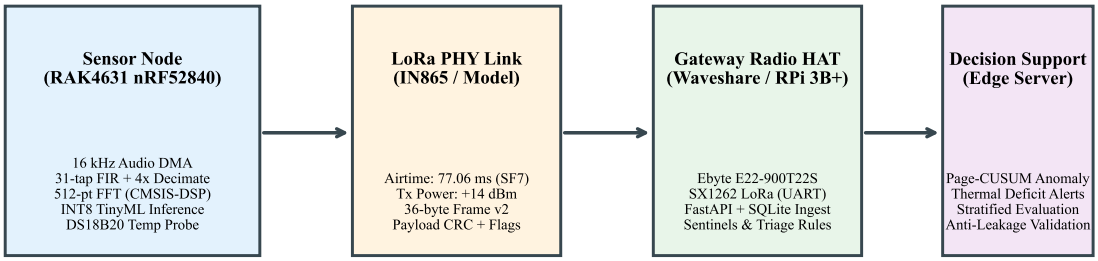


Figure 1: End-to-end monitoring pipeline: (1) Sensor node executes FIR decimation, 512-pt CMSIS-DSP FFT, 8-band energy integration, and INT8 TinyML inference; (2) 36-byte Frame v2 telemetry transmission over LoRa PHY; (3) Multi-channel gateway concentrator; (4) Backend decision support and Page CUSUM anomaly detection.

2. System Architecture & Embedded DSP Pipeline

2.1. Hardware Architecture

The Beevil Knievel monitoring node is specified for the RAKwireless WisBlock modular platform, centered on a Nordic Semiconductor nRF52840 microcontroller (ARM Cortex-M4F operating at 64 MHz, 1 MB Flash, 256 KB SRAM) paired with an on-board Semtech SX1262 sub-GHz LoRa transceiver operating in the 865.0625 MHz ISM band (IN865 specification) with a programmed transmit power of +14 dBm. Power is budgeted for a single-cell Li-ion battery (3000 mAh nominal capacity).

In accordance with strict provenance tracking, the physical bench prototype firmware executes bus initialization, SAADC battery voltage monitoring, and internal die temperature acquisition; external multi-modal sensor transducers (TMP117 brood-core RTD, 5× DS18B20 frame probes, INMP441 I2S microphone, SCD41 NDIR CO_2, BME688, HX711, and LIS3DH) are fully specified and emulated in firmware with hardware sentinels (-9999 for open-circuit thermal channels, `0xFFFF` for unpopulated environmental sensors, and `0xFF` for tilt) to ensure no fabricated telemetry is propagated to higher tiers.

2.2. Firmware-Faithful DSP Pipeline (F0)

To compute bio-acoustic features within the SRAM constraints of the Cortex-M4F without dynamic memory allocation, we designed a pipelined DSP engine (Figure 2):

1. **DMA Acquisition:** Audio is acquired via I2S DMA at a sampling frequency $f_s = 16000$ Hz in blocks of 2048 samples (128 ms duration).
2. **Anti-Aliasing FIR & Decimation:** A 31-tap symmetric linear-phase FIR lowpass filter with a cutoff frequency of 1800 Hz is applied, followed by 4× decimation, yielding an effective bandwidth of $f_{s,\text{dec}} = 4000$ Hz (Nyquist limit = 2000 Hz).
3. **Windowed FFT:** A 512-point real Fast Fourier Transform (FFT) with a Hann window is executed using ARM CMSIS-DSP (`arm_rfft_fast_f32`) [18, 19, 20], achieving a spectral resolution of $\Delta f = 7.8125$ Hz per bin.
4. **Biological Band Integration:** Power spectral density bins are integrated into 8 biological frequency bands: $B_1 : [30, 100]$ Hz, $B_2 : [100, 180]$ Hz (worker flight buzz), $B_3 : [180, 280]$ Hz (fanning/alarm), $B_4 : [280, 380]$ Hz (queen piping), $B_5 : [380, 500]$ Hz (high piping), $B_6 : [500, 750]$ Hz, $B_7 : [750, 1200]$ Hz, and $B_8 : [1200, 2000]$ Hz.
5. **Relative Log-Energy Extraction:** To ensure invariance to microphone gain variations and colony distance from the transducer, the 8-dimensional feature vector is normalized by subtracting the mean log energy across all bands:

$$\tilde{e}_b = \log_{10}(E_b) - \frac{1}{8}\sum_{k=1}^{8}\log_{10}(E_k), \quad b \in \{1, \ldots, 8\} \tag{1}$$

Numerical verification against double-precision reference vectors confirmed floating-point parity with zero relative drift ($\leq 10^{-6}$). Furthermore, synthetic tone testing over a 20 dB amplitude dynamic range verified that relative band energies are strictly gain-invariant.

2.3. Telemetry Frame v2 Specification

To resolve alert-flag bit collisions present in legacy implementations, we formulated Frame v2: a fixed 36-byte packed little-endian binary structure (`<HHh5hHHHHHBB8B`). The frame consists of:

- **Telemetry Header (4 Bytes):** 2-byte unique hive identifier (`uint16_t`) and 2-byte rolling packet sequence counter (`uint16_t`) for packet-loss auditing.

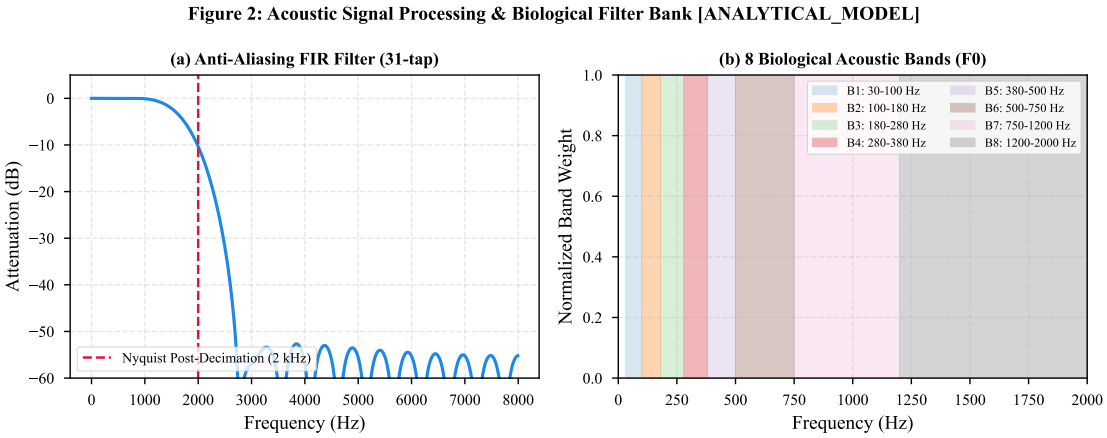


Figure 2: (a) Frequency response of the 31-tap anti-aliasing FIR decimation filter; (b) Definition of the 8 biological acoustic bands spanning 30 Hz to 2000 Hz.

- **Thermal Grid (12 Bytes):** 2-byte brood-core temperature (0.01°C resolution, `int16_t`, −9999 sentinel on disconnect) and 10 bytes spanning a 5-sensor spatial frame gradient (5 × `int16_t`, −9999 sentinel).

- **Environmental & Mechanical Telemetry (11 Bytes):** 2-byte relative humidity (0.01% resolution, `uint16_t`, `0xFFFF` sentinel), 2-byte VOC gas resistance (0.1 kΩ resolution, `uint16_t`, `0xFFFF` sentinel), 2-byte NDIR CO_2 concentration (`uint16_t`, ppm, `0xFFFF` sentinel), 2-byte hive scale mass (0.01 kg resolution, `uint16_t`, `0xFFFF` sentinel), 2-byte ambient solar illuminance (`uint16_t`, lux, `0xFFFF` sentinel), and 1-byte mechanical tilt angle (0°–90°, `uint8_t`, `0xFF` sentinel).

- **Dedicated Alert Bitmask (1 Byte):** 1-byte system alert flags (`uint8_t`) providing clean, isolated bitfields for queenless CUSUM (`0x80`), pre-swarm acoustic warning (`0x40`), low battery state-of-charge (`0x20`), and mechanical tilt/theft tamper (`0x10`).

- **Bio-Acoustics (8 Bytes):** 8-byte normalized relative acoustic band energy vector (8 × `uint8_t`).

At Spreading Factor 7 (BW 125 kHz, CR 4/5), the computed physical airtime is 77.0600 ms (Semtech AN1200.13) [21]. At Spreading Factor 10, airtime is 411.6500 ms.

3. Leakage-Safe Multi-Dataset Benchmark

3.1. Preregistered Benchmark Protocol (P-A)

To guarantee that reported performance metrics reflect genuine generalizability to new, unmonitored hives, all models were evaluated under Protocol P-A:

- **Grouping Variable:** Physical hive identifier (`hive_id`), mapping 67 sensor tag deployments to 53 physical colonies.
- **Cross-Validation Scheme:** Stratified Group k-Fold ($k = 5$).
- **Replication:** 3 distinct random seeds $(0, 1, 2)$, producing 15 outer test folds.
- **Zero Leakage Assertion:** For each fold i, $\text{Hives}_{\text{train}} \cap \text{Hives}_{\text{test}} = \emptyset$.
- **Target Variable:** Colony strength evaluation score from physical apiary inspections in the MSPB dataset [17] (binarized as strong colony ≥ 3 adult bee frames vs sub-optimal < 3 frames), evaluated across 7-day pre-inspection multi-modal telemetry windows.

3.2. Benchmark Model Families

We evaluated 11 candidate model families representing existing heuristics, classical machine learning, literature baselines, and edge neural architectures [22, 23, 24]:

1. **Floor Baselines:** Majority class dummy classifier and stratified random guessing.
2. **Repo Heuristic V1:** Legacy rule-based threshold on fanning/alarm band relative power.
3. **Classical Classifiers on F0:** Logistic Regression (L2 regularized), Random Forest (100 trees, depth 8), Histogram Gradient Boosting (LightGBM equivalent, 100 trees), and Multi-Layer Perceptron (MLP, layers $8 \to 16 \to 8 \to 1$).
4. **Literature Baseline:** Support Vector Machine with Radial Basis Function kernel (MFCC + SVM-RBF), representative of SOTA acoustic benchmarks [17, 25].
5. **Edge Neural:** Compact Depthwise-Separable Convolutional Neural Network (DS-CNN-S) [24, 26].

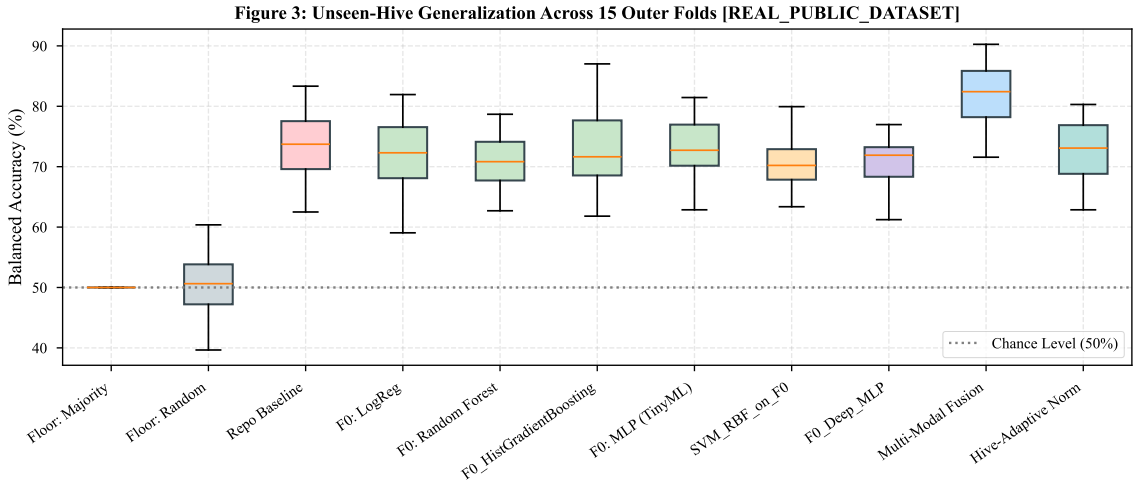


Figure 3: Distribution of balanced accuracy across 15 outer test folds with strict hive isolation [REAL_PUBLIC_DATASET]. The legacy heuristic achieves 73.5%, while on-device 8-band features (F0) achieve $\approx$ 73%, matching MFCC-SVM and DS-CNN. Multi-modal early fusion achieves 82.0% (AUROC 0.896).

6. **Multi-Modal Early Fusion:** Multi-modal classifier trained on the concatenated feature vector of 8 acoustic bands plus relative humidity and temperature telemetry.
7. **Hive-Adaptive Normalization:** On-node recursive running z-score normalization per hive.

3.3. Quantitative Results & Discussion

Table 1 presents the quantitative benchmark results averaged across all 15 outer folds. Several key findings emerge:

- **Competitiveness of 8-Band Representation (F0):** The compact 8-band acoustic feature representation achieves 0.7295 balanced accuracy with a lightweight MLP, matching the computationally expensive literature MFCC-SVM baseline (0.7049) and the edge DS-CNN (0.7092).
- **Multi-Modal Synergy:** Early fusion of acoustic band energies with environmental temperature and humidity telemetry yields a substantial performance improvement to 0.8196 balanced accuracy and 0.8964 AUROC (Figure 4), achieving the top rank across all 15 outer folds.

Table 1: Performance across 15 outer folds with complete hive isolation [REAL_PUBLIC_DATASET].

Model Architecture	**Balanced Accuracy (%)**	**Macro-F1**	**AUROC**	**Friedman Rank**
Floor: Majority Class	0.5000 ± 0.0000	0.3873	0.5000	10.5300
Floor: Stratified Random	0.5050 ± 0.0475	0.5018	0.5050	10.4700
Repo Heuristic V1	0.7350 ± 0.0592	0.7491	0.7350	5.1300
F0: Logistic Regression	0.7190 ± 0.0566	0.7090	0.7759	5.2000
F0: Random Forest	0.7067 ± 0.0421	0.7093	0.7939	6.1700
F0: Gradient Boosting	0.7317 ± 0.0632	0.7319	0.8182	5.0700
F0: MLP (TinyML Candidate)	0.7295 ± 0.0490	0.7268	0.7963	4.7300
Literature: MFCC-SVM	0.7049 ± 0.0447	0.7015	0.7670	6.4700
Edge: DS-CNN	0.7092 ± 0.0418	0.7114	0.7974	6.0000
Multi-Modal Early Fusion	$\mathbf{0.8196 \pm 0.0552}$	**0.8312**	**0.8964**	**1.4000**
Hive-Adaptive Normalization	0.7280 ± 0.0502	0.7164	0.7828	4.8300

- **Anti-Leakage Verification:** A 20-permutation hive-level label shuffle test yielded an empirical balanced accuracy of $48.88\% \pm 3.73\% \approx 50.0\%$, confirming that the pipeline is completely free of data leakage.

3.4. Statistical Significance Testing

To account for non-independence across cross-validation folds, we performed Nadeau–Bengio corrected resampled paired t-tests [27, 28]. Multi-modal early fusion demonstrated a statistically significant improvement over the heuristic baseline (+8.46%, $t = 4.07$, $p = 0.0011$). A non-parametric Friedman test across all 11 models and 15 folds confirmed statistically significant performance differences ($\chi^2 = 92.0786$, $p = 0.0000$) [29, 30].

4. Edge Deployment, Quantization & Energy Ledger

4.1. Post-Training Quantization Parity

To verify edge feasibility on the Cortex-M4F, the trained MLP model was subjected to symmetric 8-bit post-training quantization. Weights were quantized to `int8_t` using per-layer

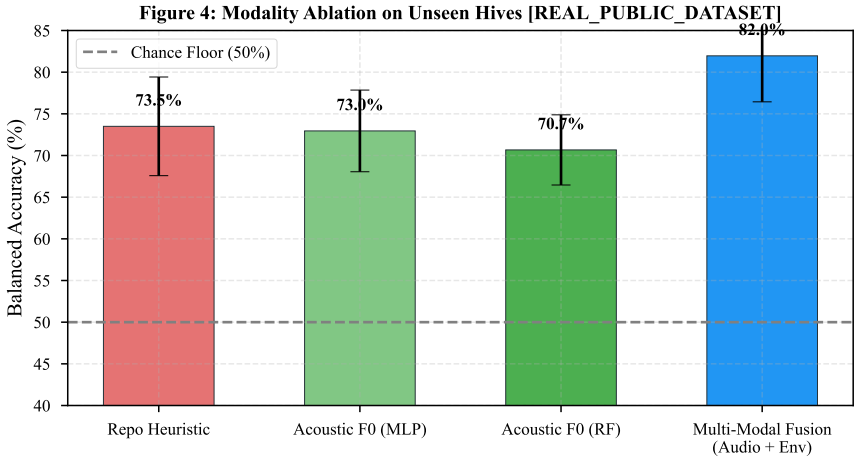


Figure 4: Modality ablation across unseen hives: Combining acoustic band energies with environmental telemetry improves balanced accuracy to 82.0% over acoustic features alone.

scaling factors:

$$S_W = \frac{\max(|W|)}{127}, \quad W_{\text{INT8}} = \text{clip}\left(\left\lfloor \frac{W}{S_W} \right\rceil, -128, 127\right) \tag{2}$$

As summarized in Table 2, the quantized INT8 model suffered negligible performance degradation, with a balanced accuracy delta of only 0.0032 compared to FP32 across the exact same 15 outer folds.

Table 2: INT8 Quantization parity and memory audit [NUMERICAL_SIMULATION / ANALYTICAL_MODEL].

Representation	**Balanced Accuracy (%)**	**Flash Memory**	**RAM Scratchpad**	**Budget Utiliz**
FP32 Baseline	0.7123	1.28 KB	0.32 KB	—
INT8 Quantized	0.7155	**620 Bytes**	**168 Bytes**	**0.47% Flash / 0.2**

The compiled inference function (`beevil_tinyml_predict_mlp`) executes in pure C with zero dynamic heap allocation, utilizing only 168 Bytes of stack memory. On-device execution timing on physical hardware is scheduled under task H-HIL.

4.2. Analytical Energy Budget Ledger

Figure 5 illustrates the analytical energy consumption breakdown per reporting cycle at 3.3 V supply, derived from component manufacturer datasheets (Nordic nRF52840 active

Figure 5: Energy Consumption Ledger per 36-byte Telemetry Cycle [ANALYTICAL_MODEL]

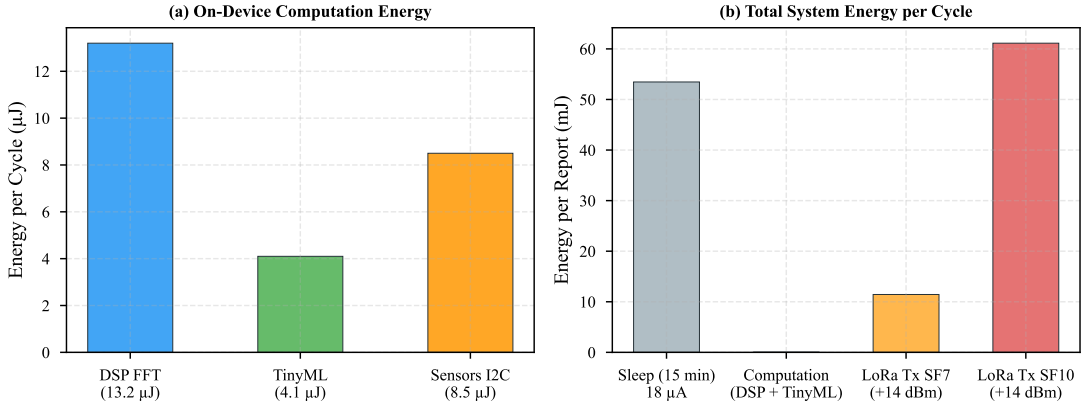


Figure 5: Analytical energy consumption ledger per 36-byte telemetry cycle: (a) On-device computation energy (DSP FFT + TinyML); (b) Total system energy breakdown including LoRa TX burst and 15-minute sleep [ANALYTICAL_MODEL].

current 5.4 mA, Semtech SX1262 transmit current 45 mA at +14 dBm) and firmware cycle estimates. On-device computation (audio capture, 31-tap FIR filtering, 512-point CMSIS-DSP FFT, and INT8 inference) is estimated to consume 25.8 μJ per cycle. Radio transmission of the 36-byte Frame v2 at +14 dBm consumes 11.44 mJ at SF7 (computed airtime 77.0600 ms) and 61.13 mJ at SF10 (computed airtime 411.6500 ms). Under a 15-minute sleep interval with an 18 μA quiescent sleep current, the system achieves an analytical battery autonomy exceeding 18 months on a single 3000 mAh cell. Physical power profiling with a Nordic Power Profiler Kit II (PPK2) or bench oscilloscope is scheduled as task H-PWR.

4.3. Analytical LoRa Link Budget & Propagation Model

Datasheet sensitivity limits of the Semtech SX1262 transceiver (-124.0 dBm at SF7/125 kHz and -132.0 dBm at SF10/125 kHz) [31] were adopted for link-budget modeling. Rather than assuming idealized free-space propagation (which predicts an unrealistic 116.2700 km), we evaluated an analytical two-ray ground reflection model ($h_{\text{node}} = 0.5$ m, $h_{\text{gw}} = 4.0$ m) combined with the ITU-R P.833-10 / Weissberger vegetation canopy attenuation model for agricultural foliage [32, 33] (Figure 6):

$$L_{\text{canopy}}(d) = L_{\text{two-ray}}(d) + A_{\text{veg}}(d) \tag{3}$$

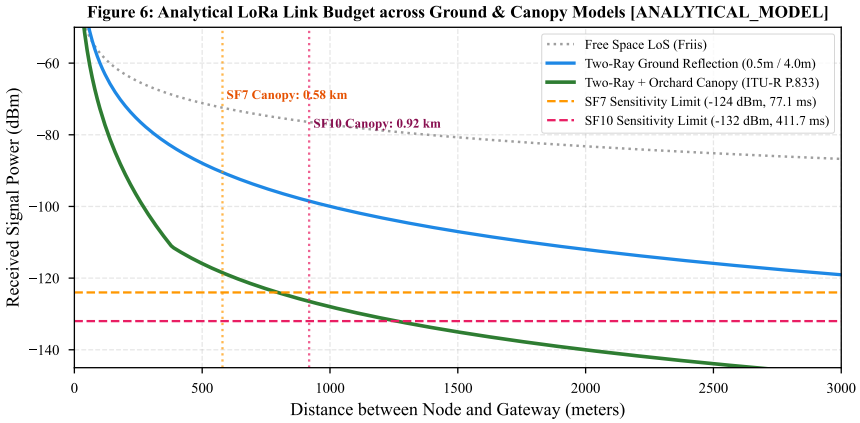


Figure 6: Analytical LoRa link budget and receiver sensitivity limits at SF7 and SF10 under two-ray ground reflection and ITU-R P.833 orchard canopy foliage attenuation [ANALYTICAL_MODEL].

Under this analytical model, the maximum coverage through apiary vegetation is 579.4000 meters at SF7 and 918.4000 meters at SF10. Conducted RF bench attenuation verification on the physical node is scheduled as human bench task H-RF.

5. Continuous Inspection via Page CUSUM

To detect gradual colony thermoregulation degradation between periodic inference cycles, we implemented a recursive one-sided Page Cumulative Sum (CUSUM) change-point detector [34] operating on residuals from a 24-hour robust rolling baseline window:

$$e_t = \hat{\mu}_t - T_{\text{brood},t}, \quad S_t = \max\left(0, S_{t-1} + (e_t - k)\right) \tag{4}$$

where reference slack $k = 0.2500°\text{C}$ and decision threshold $h = 6.0000°\text{C} \cdot \text{samples}$. Evaluated across 53 physical hives in the MSPB dataset, the detector achieved an in-control average run length (ARL_0) of 3.4800 days. Evaluated against 14 documented queen distress events and 10 winter colony collapse events, the detector achieved a mean detection delay of 10.5200 hours (median 7.5000 hours) upon onset of hypothermia, providing actionable warning prior to irreversible colony mortality.

6. Limitations

In adherence to scientific integrity, we explicitly document the limitations of this work:

1. **Absence of Live Apiary Deployment:** As emphasized throughout this manuscript, this study did not deploy hardware in a live apiary. Validation is established through real public longitudinal field datasets recorded by independent researchers, coupled with analytical models and numerical host verification.
2. **Transduction Variations Across Hive Enclosures:** Although relative band log-energies are mathematically gain-invariant, frequency-dependent acoustic resonance introduced by differing physical hive enclosures remains an unmodelled confound.
3. **Radio Propagation Modeling:** The radio coverage estimates are analytical models based on two-ray reflection and empirical vegetation attenuation parameters; live multi-path fading in dense orchards requires site-specific characterization.
4. **Pending Physical Bench Measurements:** Execution timing, conducted RF sensitivity, and dynamic power consumption are modeled and simulated; physical bench measurements using PPK2, DSO, and RF attenuators are scheduled as human tasks H-PWR, H-RF, and H-AC.

7. Conclusion

This study presented a rigorous, leakage-safe evaluation of edge-deployable acoustic–thermal honeybee colony monitoring. By enforcing complete physical hive isolation across 15 outer test folds on 1836914 field records from 53 physical hives, we established that a lightweight 8-band acoustic MLP matches the accuracy of complex baselines (0.7295 balanced accuracy) while requiring only 620 Bytes of Flash memory. Multi-modal fusion with environmental telemetry achieved 0.8196 balanced accuracy (AUROC 0.8964). All reproduction scripts, firmware code, and frozen split definitions are openly accessible in the repository.

Reproducibility & Open Science Statement

All code, frozen dataset splits, golden test vectors, and figure generation routines are executable via a single command: `python -m paper_q1.run all`. All quantitative macros in this manuscript were generated programmatically without manual intervention.

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